13.3) Partial Derivative 13|10|2026

13.3.3) - will be discussed in next ~~video~~ lecture

Main Discussion for 13.3-1 + 13.3-2

a) Partial Derivatives
   - Definition using limits
   - Interpretation of Partial Derivatives
      - Geometrical
      - Dynamical

b) Computation of Partial Derivatives
   - Using Standard Differentiation Rules

c) Higher Order Partial Derivatives

d) Can the order of differentiation be switched ?

"$\frac{\partial}{\partial x}\frac{\partial}{\partial y} = \frac{\partial}{\partial y}\frac{\partial}{\partial x}$" ?

Answer : No In general

e) Mixed Derivative Theorem
   - gives a necessary condition for "=" holds.

13.3) Partial Derivatives

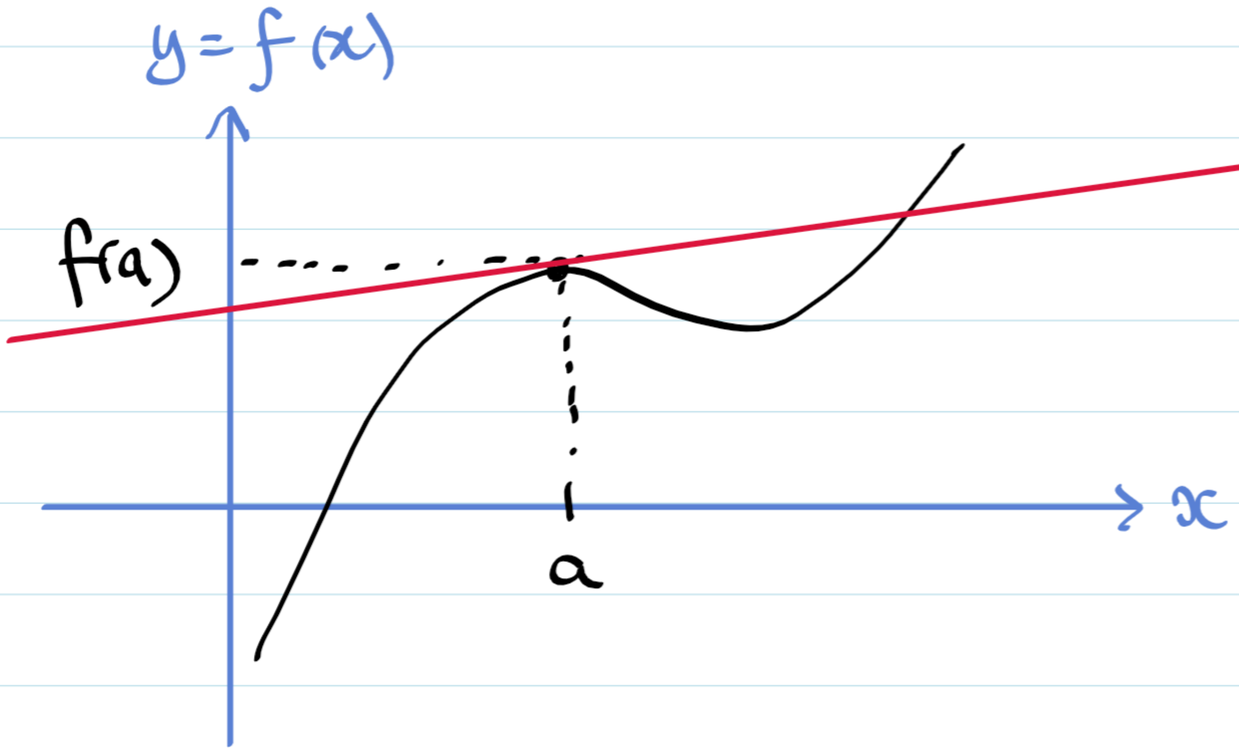
13.3.1) Partial Derivatives.

Note 13.3.1 (Differentiability in Calculus of one-variables & its interpretation). SIM1002

Let $f : \mathbb{R} \to \mathbb{R}$ and $a \in \mathbb{R}$. Then

1. the derivative of f at point a: $\quad f'(a) = \lim_{\Delta x \to 0} \dfrac{f(a+\Delta x) - f(a)}{\Delta x}$;

2. $f'(a)$ is the slope of the tangent line at point a of the graph: $y = f(x)$ in the xy-plane.



tangent line to f at point a.

$f'(a)$ = the slope of this line

$y = b$ (remain no change)

a) Defn. For $z = f(x, y)$, $(a, b) \in$ Domain of f

i) $\dfrac{\partial f}{\partial x}\begin{bmatrix} a \\ b \end{bmatrix} = \lim_{\Delta x \to 0} \dfrac{f\begin{bmatrix} a+\Delta x \\ b \end{bmatrix} - f\begin{bmatrix} a \\ b \end{bmatrix}}{\Delta x} \equiv f_x\begin{bmatrix} a \\ b \end{bmatrix}$

ii) $\dfrac{\partial f}{\partial y}\begin{bmatrix} a \\ b \end{bmatrix} = \lim_{\Delta y \to 0} \dfrac{f\begin{bmatrix} a \\ b+\Delta y \end{bmatrix} - f\begin{bmatrix} a \\ b \end{bmatrix}}{\Delta y} \equiv f_y\begin{bmatrix} a \\ b \end{bmatrix}$ or $f_y(a\ b)$

$x = a$ remain no change

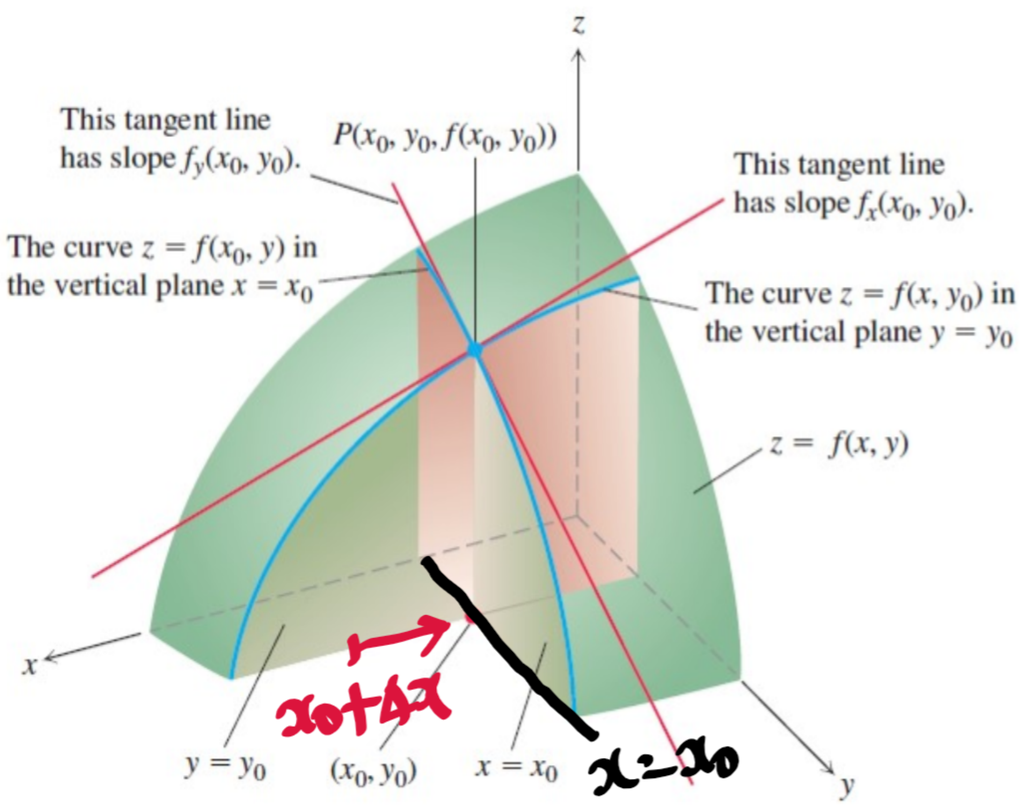


FIGURE 13.18 Figures 13.16 and 13.17 combined. The tangent lines at the point $(x_0, y_0, f(x_0, y_0))$ determine a plane that, in this picture at least, appears to be tangent to the surface.

b) Geometric Interpretation

i) $\frac{\partial f}{\partial x}\begin{bmatrix} a \\ b \end{bmatrix}$ = slope to the tangent line to $y = b$ at point (a, b)

ii) $\frac{\partial f}{\partial y}\begin{bmatrix} a \\ b \end{bmatrix}$ = slope to the tangent line to $x = a$ at point (a, b)

(#) For $y = f(x)$,

$\frac{df}{dx}$ or $\frac{dy}{dx}$ = rate of change of f wrt a unit change in x.

c) Dynamical Interpretation

i) $\frac{\partial f}{\partial x}$ = the rate of change of f per unit change in x

ii) $\frac{\partial f}{\partial y}$ = the rate of change of f per unit change in y.

d) Notation

We write $f_x \equiv \frac{\partial f}{\partial x}$ $\qquad$ $f_y \equiv \frac{\partial f}{\partial y}$

Note:

To be more precise we should write down the corresponding point in the Domain as well.

(eg) $f_x(3\ 5)$ or $\frac{\partial f}{\partial x}(3\ 5)$ or $\left.\frac{\partial f}{\partial x}\right|_{\substack{x=3 \\ y=5}}$

But in most of the time, the corresponding point will be omitted to simplify the expression $\left(f_x, \frac{\partial f}{\partial x}\right)$

e) How to compute f_x, f_y ?

Remark 13.3.1 (Rule for finding Partial Derivatives).

1. To find f_x: regard y as constant and differentiate $f(x,y)$ wrt. x.
2. To find f_y: regard x as constant and differentiate $f(x,y)$ wrt. y.

recall: $\frac{d}{du}\tan^{-1}u = \frac{1}{1+u^2}$

Eg 13.3.12) $f = \tan^{-1}\left(\frac{y}{x}\right)$

Soln

$$f_x = \frac{\partial}{\partial x}\left[\tan^{-1}\left(\frac{y}{x}\right)\right]$$

treat y = const

$$= \frac{1}{1+\left(\frac{y}{x}\right)^2}\,\frac{\partial}{\partial x}\left(\frac{y}{x}\right)$$

$$= \frac{1}{1+\left(\frac{y}{x}\right)^2}\, y\left(-\frac{1}{x^2}\right) = ____$$

$$f_y = \frac{\partial}{\partial y}\left[\tan^{-1}\left(\frac{y}{x}\right)\right]$$

consider x as a const

$$= \frac{1}{1+\left(\frac{y}{x}\right)^2}\,\frac{1}{x}$$

Eg 13.3.30) $f = yz\ln(xy)$

Soln:

a) $\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}\left[yz\ln(xy)\right]$ - consider y, z as const then diff wrt x

$= yz \cdot \frac{y}{xy}$

$= \frac{yz}{x}$

c) $\frac{\partial f}{\partial z} = \frac{\partial}{\partial z}\left[yz\ln(xy)\right]$ - consider x, y as const then diff wrt z

$= y \cdot 1 \cdot \ln(xy)$

$= y\ln(xy)$

b) $\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}\left[yz\ln(xy)\right]$ - consider x, z as const then diff wrt y

$= 1 \cdot z\ln(xy) + yz \cdot \frac{x}{xy}$

$= z\ln(xy) + z$

Example (13.3.75). [(13.3.69) in 15th Ed.]

Find the value of $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ at the point $(1,1,1)$ if the equation

$$xy + z^3x - 2yz = 0$$

defines z as a function of two independent variables x and y.

$\frac{\partial x}{\partial y} = 0$, $\frac{\partial y}{\partial x} = 0$

Soln

a) $\frac{\partial}{\partial x}(0) = \frac{\partial}{\partial x}(xy + z^3x - 2yz)$

$$0 = 1\cdot y + 3z^2\frac{\partial z}{\partial x}\cdot x + z^3\cdot 1 - 2y\frac{\partial z}{\partial x}$$

$$0 = y + z^3 + (3z^2 - 2y)\frac{\partial z}{\partial x}$$

when $x=1$, $y=1$, $z=1$, we have

$$0 = 1 + 1 + (3-2)\frac{\partial z}{\partial x}$$

$\therefore \frac{\partial z}{\partial x} = -2$ [To be more precise: write $\frac{\partial z}{\partial x}(1,1) = -2$.]

(b) $\frac{\partial}{\partial y}0 = \frac{\partial}{\partial y}(xy + z^3x - 2yz)$

$$0 = x + 3z^2\frac{\partial z}{\partial y} - 2z - 2y\frac{\partial z}{\partial y}$$

when $x=1$, $y=1$, $z=1$, we have

$$\frac{\partial z}{\partial y} = 1$$

Example (13.3.68.). [(13.3.76) in 15th Ed.]
Let $f(x,y) = x^2 + y^3$. Find the slope of the line tangent to this surface at the point $(-1, 1)$ and lying in the

(a) plane $x = -1$, (b) plane $y = 1$

$x = \text{const}$ $y = \text{const}$

Find $f_y\begin{bmatrix}-1\\1\end{bmatrix}$ Find $f_x\begin{bmatrix}-1\\1\end{bmatrix}$

Soln

a) The slope ..... point $(-1,1)$ in the plane $x=-1$:

$$f_y\begin{bmatrix}-1\\1\end{bmatrix} = \frac{\partial}{\partial y}(x^2+y^3)\Big|_{\substack{x=-1\\y=1}}$$

$$= (0 + 3y^2)\Big|_{\substack{x=-1\\y=1}} = 3$$

b) The slope ..... point $(-1,1)$ in the plane $y=1$:

$$f_x\begin{bmatrix}1\\-1\end{bmatrix} = \frac{\partial}{\partial x}(x^2+y^3)\Big|_{\substack{x=-1\\y=1}}$$

$$= (2x + 0)\Big|_{\substack{x=-1\\y=1}}$$

$$= -2.$$

f) when to use "d" and when for "∂" ?

(i) We use "d" when there is only ONE parameter

(ii) We use "∂" when there are TWO or MORE parameters

(eg) $y = f(t)$: use $\frac{df}{dt}$ or $\frac{dy}{dt}$

$y = \sin(t^2) \Rightarrow \frac{dy}{dt} = \cos(t^2) \cdot 2t$

$w = f(s, x, \theta)$ use $\frac{\partial w}{\partial s}$, $\frac{\partial w}{\partial x}$ $\frac{\partial w}{\partial \theta}$

$w = s^2 \cos\theta + x^3 \sin\theta$

$\frac{\partial w}{\partial s} = 2s\cos\theta$, $\frac{\partial w}{\partial x} = 3x^2 \sin\theta$

$\frac{\partial w}{\partial \theta} = -s^2 \sin\theta + x^3 \cos\theta$.

13.3.2) Higher Order Partial Derivative.

a) There is a concept of "higer order derivatives" as in SIM1002.

Notation.

i) $f_x = \frac{\partial f}{\partial x}$ $f_y = \frac{\partial f}{\partial y}$ $f_z = \frac{\partial f}{\partial z}$ ...

ii) $f_{xx} = \frac{\partial}{\partial x}\frac{\partial}{\partial x} f$ $f_{yx} = \frac{\partial}{\partial x}\frac{\partial f}{\partial y}$ $f_{zy} = \frac{\partial}{\partial y}\frac{\partial}{\partial z} f$ etc

iii) $f_{xyy} = \frac{\partial}{\partial y}\frac{\partial}{\partial y}\frac{\partial}{\partial x} f$ $f_{xyzx} = \frac{\partial}{\partial x}\frac{\partial}{\partial t}\frac{\partial}{\partial y}\frac{\partial}{\partial x} f$

Eg (extra) $f = xy\ln(z+x^2)$

$$f_x = y\ln(z+x^2) + \frac{2x^2y}{z+x^2} = \frac{\partial f}{\partial x}$$

$$f_{xz} = \frac{y}{z+x^2} - \frac{2x^2y}{(z+x^2)^2} = \frac{\partial}{\partial z}\frac{\partial}{\partial x} f$$

$$f_{xzy} = \frac{1}{z+x^2} - \frac{2x^2}{(z+x^2)^2} = \frac{\partial}{\partial y}\frac{\partial}{\partial z}\frac{\partial}{\partial x} f$$

$\therefore$

Example. Show that the following function is a solution of the wave equation $w_{tt} = c^2 w_{xx}$.

13.3.93. $w = \sin(x+ct) + \cos(2x+2ct)$.

Soln

$w_t = c\cos(x+ct) - 2c\sin(2x+2ct)$

$w_{tt} = -c^2\sin(x+ct) - 4c^2\cos(2x+2ct)$

$= c^2\{-\sin(x+ct) - 4\cos(2x+2ct)\}$ — (1)

$w_x = \cos(x+ct) - 2\sin(2x+2ct)$

$w_{xx} = -\sin(x+ct) - 4\cos(2x+2ct)$ — (2)

Subst (2) in (1),

$w_{tt} = c^2 w_{xx}$

b) *Note* 13.3.2. In general, $f_{xy} \neq f_{yx}$.

Nevertheless $f_{xy} = f_{yx}$ for most of the cases.

Theorem (pp.783: The Mixed Derivative Theorem). [pp.795 in 15th Ed.]
Let $\mathscr{D}$ be an open set on $\mathbb{R}^2$ and let $f : \mathscr{D} \subset \mathbb{R}^2 \to \mathbb{R}$. If the functions f_{xy}, f_{yx} are continuous on $\mathscr{D}$, then $f_{xy} = f_{yx}$ on $\mathscr{D}$.

Proof. (See Textbook, pp. AP-49) □

Rmk:

Since most of the functions f we studied satisfying the conditions f_{xy} and f_{yx} both exist and continuous,

we have $f_{xy} = f_{yx}$ (for most of the cases!)

Example (13.3.82). Let $f(x,y) = \begin{cases} xy\dfrac{x^2-y^2}{x^2+y^2}; & (x,y) \neq (0,0) \\ 0; & (x,y) = (0,0) \end{cases}$.

(a) Show that $f_y(x,0) = x$ for all x and $f_x(0,y) = -y$ for all y.

(b) Show that $f_{xy}(0,0) = -1$ and $f_{yx}(0,0) = 1$. $\Rightarrow$ $f_{xy}\begin{bmatrix}0\\0\end{bmatrix} \neq f_{yx}\begin{bmatrix}0\\0\end{bmatrix}$!

Rmk : Since f is a piecewise function we need to use the defn to compute $f_x(0\,0)$, $f_y(0\,0)$

a) $$\frac{\partial f}{\partial y}\begin{bmatrix}x\\0\end{bmatrix} = \lim_{k\to 0} \frac{f\begin{bmatrix}x\\0+k\end{bmatrix} - f\begin{bmatrix}x\\0\end{bmatrix}}{k}$$

$$= \lim_{k\to 0} \frac{xk\,\dfrac{x^2-k^2}{x^2+k^2} - 0}{k}$$

$$= x\,\frac{x-0}{x^2+0} = x$$

b) $$\frac{\partial f}{\partial x}\begin{bmatrix}0\\y\end{bmatrix} = \lim_{h\to 0} \frac{f\begin{bmatrix}0+h\\y\end{bmatrix} - f\begin{bmatrix}0\\y\end{bmatrix}}{h}$$

$$= \lim_{h\to 0} \frac{hy\,\dfrac{h^2-y^2}{h^2+y^2} - 0}{h}$$

$$= y\,\frac{0-y^2}{0+y^2} = -y$$

Example (13.3.82). Let $f(x,y) = \begin{cases} xy\dfrac{x^2-y^2}{x^2+y^2}; & (x,y) \neq (0,0) \\ 0; & (x,y) = (0,0) \end{cases}$.

(a) Show that $f_y(x,0) = x$ for all x and $f_x(0,y) = -y$ for all y.

(b) Show that $f_{xy}(0,0) = -1$ and $f_{yx}(0,0) = 1$.

Solution [Cont]

b) From (a) $f_y\begin{bmatrix} x \\ 0 \end{bmatrix} = x$, $f_x\begin{bmatrix} 0 \\ y \end{bmatrix} = -y$

$$f_{xy}\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \frac{\partial}{\partial y} f_x\begin{bmatrix} 0 \\ y \end{bmatrix}\Big|_{y=0} = \frac{\partial}{\partial y}(-y)\Big|_{y=0} = -1$$

$$f_{yx}\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \frac{\partial}{\partial x}\Big| f_y\begin{bmatrix} x \\ 0 \end{bmatrix}_{x=0} = \frac{\partial}{\partial x}\Big|_{x=0}(x) = +1$$

Rmk :

(i) In this example, you can verify

$$f_{xy} = \frac{x^6 + 9x^4y^2 - 4x^2y^4 + 4y^6}{(x^2+y^2)^3} \quad \text{for } (x,y) \neq (0,0)$$

(ii) you can also verify that (using 2 path method In Sect 13.2)

$\lim_{\substack{x\to 0 \\ y\to 0}} f(x,y)$ does not exist !

$\therefore$ f_{xy} is not continuous at $(0,0)$!